Neutron Kinetics with Multiplication

NE 630 | Nuclear Reactor Theory

26 LESSON

Read: FNRP 5.2–5.3.

Students will be able to write down, solve, and explain each term of a first-order, ordinary differential equation that models the neutron flux, the neutron density, or the number of neutrons in a multiplying system as a function of time.

1 Review: Kinetics without Multiplication

By first eliminating space from the diffusion equation and then assuming $\phi(E,t) = \phi(t)\varphi(E)$ for some known $\varphi(E)$, we found

$$\frac{1}{\bar{v}}\frac{d\phi}{dt} + \bar{\Sigma}_t\phi(t) = \bar{\Sigma}_s\phi(t) + \frac{1}{k_{\text{eff}}}\bar{\nu}\bar{\Sigma}_f\phi(t) + S(t)\,, \qquad (25.1 \to 25.3)$$

or, with $\phi = n'''\bar{v}$,

$$\frac{dn'''}{dt} + \left(\frac{1}{l_\infty}\right)n'''(t) = S(t)\,, \qquad \text{(Like FNRP 5.2 and 5.7)}$$

where the mean neutron speed and lifetime are

$$\bar{v} = \frac{\displaystyle\int_0^\infty \varphi(E)\,dE}{\displaystyle\int_0^\infty \frac{\varphi(E)}{v(E)}\,dE} \quad \text{and} \quad l_\infty = \frac{1}{\bar{v}\bar{\Sigma}_a}\,,$$

and the neutron speed is $v(E) \approx 1.384 \cdot 10^6\sqrt{E}$ cm s^{-1} for E in eV.

2 Adding Back Multiplication

With multiplication, we have[1]

$$\frac{1}{\bar{v}}\frac{d\phi}{dt} + \bar{\Sigma}_t\phi(t) = \bar{\Sigma}_s\phi(t) + \bar{\nu}\bar{\Sigma}_f\phi(t) + S(t)\,. \qquad (26.1)$$

Let $\bar{\Sigma}_a = \bar{\Sigma}_t - \bar{\Sigma}_s$ and $\phi(t) = n'''(t)\bar{v}$ in (26.1). Then, multiply by volume so that

$$\begin{aligned}\frac{dn}{dt} &= \bar{v}(\bar{\nu}\bar{\Sigma}_f - \bar{\Sigma}_a)n(t) + S(t)\\ &= \bar{\Sigma}_a\bar{v}\left(\frac{\bar{\nu}\bar{\Sigma}_f}{\bar{\Sigma}_a} - 1\right)n(t) + S(t)\end{aligned}$$

$$\boxed{\frac{dn}{dt} = \left(\frac{k_\infty - 1}{l_\infty}\right)n(t) + S(t)}\,. \qquad \text{(FNRP 5.12)}$$

Here and below, $n(t)$ is the total neutron population, and $S(t)$ denotes the volume-integrated source in neutrons/s (rather than neutrons/cm^3/s).

3 Kinetics with Multiplication

For finite systems with leakage, the text explains how to incorporate the non-leakage probability P_{NL}, and the resulting FNRP (5.22) is identical FNRP (5.12) without the ∞'s on k and l:

$$\frac{dn}{dt} = \left(\frac{k-1}{l}\right)n(t) + S(t) \qquad \text{(FNRP 5.22)}$$

Example: Let $S(t) = 0$ and $n(0) = n_0$. Determine $n(t)$.

Ans. The equation is separable, i.e., $dn/n = [(k-1)/l]dt$, and direct integration yields

$$\boxed{n(t) = n_0 e^{\left(\frac{k-1}{l}\right)t}}\,, \qquad \text{(FNRP 5.24)}$$

or, recalling, $\rho = (k-1)/k$,

$$\boxed{n(t) = n_0 e^{(\rho/\Lambda)t}}\,, \qquad \text{(FNRP 5.24)}$$

where $\Lambda = l/k$ is the **prompt generation time**.

4 Kinetics of Source-Driven, Subcritical Systems

When $S(t) \neq 0$, we go from

$$\frac{d}{dt}\left(n(t)e^{-\left(\frac{k-1}{l}\right)t}\right) = S(t)e^{-\left(\frac{k-1}{l}\right)t}.$$

to

$$n(t') = \frac{\displaystyle\int_0^{t'}\left(S(t)e^{-\left(\frac{k-1}{l}\right)t}\right)dt + n(0)}{e^{-\left(\frac{k-1}{l}\right)t'}}$$

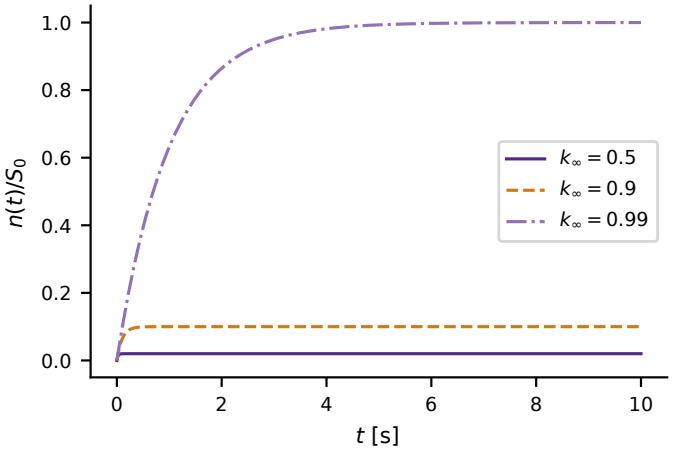
If $S(t) = S_0$, then

$$n(t) = \frac{lS_0}{k-1}\left(e^{\frac{k-1}{l}t} - 1\right) + n(0)e^{\frac{k-1}{l}t}\,.$$

If we further set $n(0) = 0$, we get FNRP (5.26).

5 Revisiting Subcritical Multiplication

Let $n(0) = 0$, $S_0 = 1$, and $l_\infty = 10^{-2}$. What happens to $n(t)$ for $0 \le t \le 10$ s as k_∞ increases from 0 to 1?[2]



Ponderable: How would the curve look when k_∞ is exactly 1?

[1] For source-free, *steady-state* problems, the multiplication factor k_{eff} allowed us to adjust the fission rate and enforce a steady state solution. For kinetics problems, we need no such adjustment!

[2] Recall that, for an infinite, homogeneous medium, $\phi = S_0/(\bar{\Sigma}_a - \bar{\nu}\bar{\Sigma}_f) = (S_0/\bar{\Sigma}_a)/(1-k_\infty)$!

6 Appendix: Solving the ODE

Problem. Let $S(t) = 0$ and $n(0) = n_0$. Determine $n(t)$.

1. Write Down the Equations

$$\frac{dn}{dt} = \left(\frac{k-1}{l}\right) n(t), \quad n(0) = n_0 .$$

2. Identify Type and Familiar Form

This first-order, initial-value problem is **separable**, and we can write

$$\frac{dn}{n(t)} = \left(\frac{k-1}{l}\right) dt .$$

3. Integrate Using Appropriate Technique

$$\int_{n(0)}^{n(t)} \frac{dn'}{n'} = \left(\frac{k-1}{l}\right) \int_0^t dt'$$

$$\ln\left(\frac{n(t)}{n(0)}\right) = \left(\frac{k-1}{l}\right) t$$

$$\Rightarrow \quad n(t) = n(0)\, e^{\left(\frac{k-1}{l}\right)t}$$

4. Apply Initial Conditions

$$\boxed{n(t) = n_0 e^{\left(\frac{k-1}{l}\right)t}}$$

With a source: For constant $k < 1$, let $\tau = l/(1-k)$. The ODE is linear; multiply by the integrating factor $e^{t/\tau}$:

$$\frac{dn}{dt} + \frac{n}{\tau} = S(t) \quad \Longrightarrow \quad \frac{d}{dt}\left(ne^{t/\tau}\right) = S(t)e^{t/\tau}.$$

Thus, for any specified source history,

$$\boxed{n(t) = n(0)e^{-t/\tau} + \int_0^t S(t')e^{-(t-t')/\tau}\, dt'.}$$

For a piecewise source, solve one interval at a time and use the final population from one interval as the initial condition for the next.

7 An Accelerator-Driven Source

An accelerator beam strikes a target, producing neutrons that enter a subcritical assembly. At fixed beam energy, target conditions, and coupling, assume $S(t)$ is proportional to the beam current. Pulsed targets are used in practice as neutron sources (e.g., SNS), but accelerator driven, sub-critical reactors remain an active area of research.

Assumptions: Fixed spatial shape and spectrum; constant k and l; no feedback or depletion. Treat all fission neutrons as prompt, as in this lesson. Delayed-neutron precursors and their slow post-pulse tail are omitted. Use

$$k = 0.90, \qquad l = 10^{-4}\ \text{s}, \qquad S_0 = 10^8\ \text{neutrons/s}.$$

Then $\tau = 1$ ms and the steady population for $S = S_0$ is $n_{ss} = S_0\tau = 10^5$ neutrons. Start each exercise with $n(0) = 0$.

8 Time-Dependent Sources: ODE Practice

Example 1: One beam pulse. Let $S(t) = S_0$ for $0 \le t < T$, then turn the beam off, with $T = 2$ ms. Determine $n(t)$ during and after the pulse. Is $n(t)$ continuous at T? Is dn/dt?

Ans. Solve the constant-source problem, then use $n(T)$ as the initial condition for the source-free decay:

$$n(t) = \begin{cases} S_0\tau(1 - e^{-t/\tau}), & 0 \le t \le T, \\ S_0\tau(1 - e^{-T/\tau})e^{-(t-T)/\tau}, & t > T. \end{cases}$$

The population is continuous, but its derivative jumps by $-S_0$ at beam shutoff. Check: $n(T) = 8.65 \times 10^4$ and $n(T + 3\tau) = 4.30 \times 10^3$ neutrons.

Example 2: A gradual beam startup. Ramp the beam current from zero to its full value over $T = 2$ ms:

$$S(t) = \begin{cases} S_0 t/T, & 0 \le t \le T, \\ S_0, & t > T. \end{cases}$$

Determine $n(t)$ on both intervals. Compare $n(T)$ with Example 1. Does reaching full beam current imply that the neutron population is already at steady state?

Check. During the ramp,

$$n(t) = \frac{S_0\tau}{T}\left[t - \tau(1 - e^{-t/\tau})\right].$$

At T, $n(T) = 5.68 \times 10^4$ neutrons. Thereafter, the difference $n(t) - n_{ss}$ decays exponentially.

Example 3: A fading beam. Let $S(t) = S_0 e^{-t/T_s}$. Obtain $n(t)$ for $T_s \neq \tau$. Then solve the special case $T_s = \tau$ directly, or take the limit of your result. For $T_s = 2\tau$, when is the neutron population largest? Why is the maximum not at $t = 0$?

Check. When $T_s = \tau$, $n(t) = S_0 t e^{-t/\tau}$. When $T_s = 2\tau$, the peak is at $t = 2\tau \ln 2$ and $n_{max} = S_0\tau/2 = 5.00 \times 10^4$ neutrons.

Example 4: A modulated beam. Let $S(t) = S_0[1 + \epsilon \cos(\omega t)]$, with $0 < \epsilon < 1$. Find the long-time periodic response in the form

$$n_{per}(t) = S_0\tau + A\cos(\omega t - \delta).$$

Determine A and δ, then add the transient needed for $n(0) = 0$. For $\epsilon = 0.20$, compare $\omega\tau = 0.1$, 1, and 10. Does the neutron population follow the rapid beam modulation?

Check. $A/(S_0\tau) = \epsilon/\sqrt{1 + (\omega\tau)^2}$ and $\delta = \tan^{-1}(\omega\tau)$. The three relative amplitudes are 0.199, 0.141, and 0.0199.